
Inferring Spatial Point Processes from a Single Realization with Topological Deep Learning

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Abstract

Spatial point processes (SPPs) are widely used to model data consisting of discrete event locations. A range of statistical models has been developed to capture spatial interactions, from homogeneous Poisson processes describing complete spatial independence to models capturing clustering, repulsion, and more complex spatial dependence. In many applications, only a single point pattern is available, and we seek a plausible generative model from which additional patterns with a similar underlying distribution can be simulated. This requires selecting an appropriate model family and estimating its parameters from a single realization. We develop a unified pipeline for these tasks across multiple SPP families, in which neural networks trained on simulated patterns combine classical spatial summary statistics with topological features derived from persistent homology. We show that these features improve estimation performance, with the resulting pipeline achieving state-of-the-art results on classical benchmarks. In addition, we analyze the state space of the classification problem, characterizing regimes in which different point process families are indistinguishable. Nevertheless, we show that even in these ambiguous regimes, our pipeline can generate new realizations that preserve the statistical properties of the observed pattern. Together, these results provide a practical framework for fitting spatial models and simulating patterns consistent with the observed data.

1 INTRODUCTION

- **Setting:** a single observed pattern; we want a plausible generative model from an established parametric family, from which similar patterns can be simulated. This means selecting the family and estimating its parameters from one realization.
- **Hook:** classification accuracy over the 8 families looks weak (0.45; 0.75 in the structured regime), yet the end-to-end skill of the fitted models is high (0.90, against 0.74 for minimum contrast, whose gap comes from the two families K cannot fit). Classification errors cost no fidelity: skill 0.90 with the predicted family vs 0.89 with the correct one.
- **Why accuracy is the wrong target near CSR:** every model struggles on patterns near complete spatial randomness, and any prior that is not cherry-picked produces many of them. This is a property of the problem, not a failure of the models. What matters is whether the fitted model reproduces the pattern.
- **Contributions:**
 1. **Scoring** (Section 5.1): a fit is scored by how well its simulations match an independent held-out replicate, with a strictly proper kernel score on local configurations, reported as skill between CSR and the true model.
 2. **Regime boundary** (Section 6): for every family except cell, detectability from CSR is governed by one physical coordinate (cluster overlap ω or a pair-count signal-to-noise ratio ζ) times a fitted power of $\bar{n}$. Before this boundary even the oracle does not beat Poisson.
 3. **Pipeline** (Sections 4 and 6): classifier then family-specific estimator, $p(f, \theta \mid x) = p(f \mid x)p(\theta \mid x, f)$. Structured patterns sent to Poisson cost nothing, wrong-family fits are

about as good as right-family ones. The pipeline beats minimum contrast at selection; end to end, it matches minimum contrast given the true family where K can be fitted, and fits Strauss and cell, where it cannot.

4. **TDA features** (Section 6): in the structured regime, persistent homology cuts estimation error by 26% on Strauss and 18% on cell, the two families minimum contrast cannot fit (cell has $K(r) = \pi r^2$ exactly); alpha carries Strauss and DTM₁₀ cell.

2 BACKGROUND

We define notation, review the summary statistics that serve as ParamNet’s classical features, and introduce the eight families we consider. A spatial point process X is a random collection of points in $\mathbb{R}^n$. We focus on single realizations of X restricted to $W \subseteq \mathbb{R}^2$, i.e. $\mathbf{x} = X \cap W$. We write $\mathbf{x} := \{x_1, \dots, x_{N(W)}\}$, where the number of points $N(W)$ is random with $\mathbb{E}N(W) = \bar{n}$.

Summary statistics. Second-order and nearest-neighbor summary statistics are widely used to analyze SPPs. We consider the K , L , F , G , and J functions defined for a stationary process with intensity λ . Ripley’s K function is defined so that $\lambda K(r)$ is the expected number of further points within distance r of a typical point in X . The L function,

$$L(r) = \sqrt{K(r)/\pi},$$

rescales K to units of distance, and is more widely used in practice. For small r , $L(r) - r > 0$ can suggest clustering, and $L(r) - r < 0$ suggests regularity.

The remaining functions are based on distances to the nearest point of X . The empty-space function

$$F(r) = \mathbb{P}(d(u, X) \leq r)$$

is the distribution of the distance from an arbitrary fixed location $u \in \mathbb{R}^2$ to the nearest point of X . The nearest-neighbor function

$$G(r) = \mathbb{P}(d(x, X \setminus \{x\}) \leq r)$$

is the distribution of the distance from a typical point x of X to its nearest neighbour. Intuitively, F measures the size of the gaps between points, while G measures how tightly points are packed around each other. The J function combines them,

$$J(r) = \frac{1 - G(r)}{1 - F(r)}, \quad F(r) < 1,$$

and compares what a typical point sees with what an arbitrary location sees: $J(r) < 1$ indicates that points are closer to each other than arbitrary locations are to points (clustering), and $J(r) > 1$ indicates the opposite (regularity).

Table 1: The eight families, grouped by second-order structure, with their parameters and the regime coordinate η that governs detectability (Section 6). ω : cluster overlap, the cluster scale in units of the parent spacing (ω_{in} : of the inner level of nested); ζ : a pair-count signal-to-noise ratio, for the regular families the expected number of missing close pairs. Patterns approach CSR as ω grows or ζ shrinks. Every family except cell has one such coordinate; cell matches CSR at second order.

Family	Parameters	Regime coordinate
Poisson (CSR)	$\bar{n}$	—
<i>Clustered</i>		
Thomas	κ, μ, σ	$\omega = \sigma\sqrt{\kappa}$
Nested	$\kappa, \mu_1, \mu_2, \sigma_1, \sigma_2$	$\omega_{\text{in}} = \sigma_2\sqrt{\kappa\mu_1}$
Ring	$\kappa, \mu, \rho, \sigma$	$\omega = \rho\sqrt{\kappa}$
LGCP	$\mu_{\log}, \sigma^2, s$	$\zeta = (e^{\sigma^2} - 1)s\bar{n}$
<i>Regular</i>		
Matérn II	R, λ_p	$\zeta = R\bar{n}$
Strauss	$\bar{n}, q, R$	$\zeta = qR\bar{n}$
<i>Second-order CSR</i>		
Cell	$\bar{n}, k$	none (U-shaped in k)

Families. We consider eight families of stationary SPPs (Table 1, Figure 1; definitions in Appendix A). The homogeneous Poisson process serves as the model for *complete spatial randomness* (CSR). We include four *clustered* families: three Neyman–Scott processes, in which Poisson parents generate offspring displaced by a kernel (Thomas, nested Thomas, and ring), and the log-Gaussian Cox process (LGCP), whose intensity is a random field. Additionally, we consider two *regular* families: Matérn II, which thins a Poisson process to enforce a hard-core distance R , and Strauss, which penalizes (rather than forbids) pairs closer than R . Finally, the Baddeley–Silverman cell process has $K(r) = \pi r^2$ exactly, so it is indistinguishable from CSR at second order despite not being Poisson; this motivates the use of topological features (Section 4).

Every family except cell approaches CSR along a single regime coordinate (Table 1; Section 6): for the Neyman–Scott families the cluster overlap $\omega = \sigma\sqrt{\kappa}$ represents the cluster scale in units of the parent spacing. We obtain crisp clusters for $\omega \ll 1$ and approach CSR as $\omega \rightarrow \infty$. Furthermore, some non-Poisson families approach one another over part of their respective parameter spaces. [Notably, Thomas, nested and ring \(Appendix A\).](#)

3 DATA GENERATION

We train and evaluate on synthetic patterns, simulated with standard algorithms, so that the true fam-

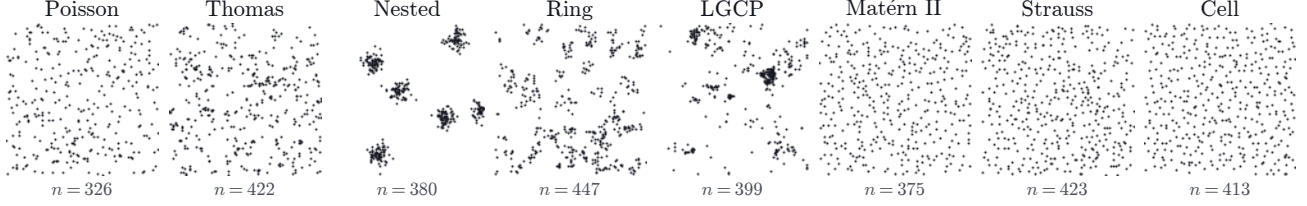


Figure 1: Example realizations on $W = [0, 1]^2$ of the eight families in structured regimes. n : number of points.

ily and parameters are known. For each family f we construct a prior π_f over its parameter space. We then draw parameters $\theta \sim \pi_f$ and simulate two independent replicates on $W = [0, 1]^2$ for each (f, θ) . The first realization, $\mathbf{x}_{f,\theta}$, acts as the observed pattern and is the input to the pipeline. The second realization, $\mathbf{y}_{f,\theta}$, is held out and used to score a model’s fit (Section 5.1).

Priors. Each θ is drawn in three steps. First, the expected number of points in W is drawn as $\bar{n} \sim \text{log-uniform}[50, 950]$, this is the same for every family. Second, the remaining degrees of freedom are drawn log-uniformly in scale-free units, for example the overlap ω for the cluster families and $R\sqrt{\bar{n}}$ for the hard-core families. Each range extends from strongly structured patterns to pattern indistinguishable from CSR. Finally, the last parameter is solved so that $\mathbb{E}N(W) = \bar{n}$ (e.g. $\kappa = \bar{n}/\mu$ for Thomas). The only draws excluded are degenerate cluster patterns, with fewer than four effective clusters in W . [We deliberately keep our priors as wide and un-curated as possible to reflect ...](#) Ranges and simulator setting as in Appendix B.

Size and splits. [For each family, we draw 50 000 parameter vectors with two replicates each, 100 000 patterns per family and 800 000 in total.](#) The data is split by θ into train, test, and validation blocks of 37 000, 12 000, and 1 000 parameter vectors, respectively. [Both replicates of each \$\theta\$ fall in the same block, so no parameter setting seen in training appears in the test block.](#) See Appendix B for details on sweep ranges, constraints and simulator settings.

4 PARAMNET

Given an input pattern $\mathbf{x}_{f,\theta}$, our pipeline produces a fit $\hat{P} = (\hat{f}, \hat{\theta})$ via a two-step procedure,

$$p(f, \theta \mid \mathbf{x}_{f,\theta}) = p(f \mid \mathbf{x}_{f,\theta}) p(\theta \mid \mathbf{x}_{f,\theta}, f).$$

An 8-way classifier predicts $\hat{f}$, followed by a family-specific estimator that produces $\hat{\theta}$. In place of a learned estimator, patterns classified as Poisson automatically receive $\hat{\lambda} = n$, which is the exact maximum-likelihood estimate.

Both stages of *ParamNet* share a single architecture, trained once as the classifier and once (per family) as an estimator (Figure 2). ParamNet receives two descriptions of $\mathbf{x}_{f,\theta}$: classical features derived from second-order summaries, and topological features obtained by vectorizing persistence diagrams. Each feature type passes through its own convolutional encoder, and the resulting embeddings are concatenated with $\log n$ before a shared FCNN head.

Classical features. Let $\alpha(r)$ be a second-order summary function in $\{L(r) - r, F(r), G(r), J(r)\}$. We compute each of these functions at 512 linearly-spaced radii in the interval $(0, 0.25]$. For each function, we obtain a feature vector $[\alpha(r_1), \dots, \alpha(r_{512})] \in \mathbb{R}^{512}$. [For \$L\$ we apply Ripley’s isotropic edge correction; for \$F\$ and \$G\$ we apply the border \(reduced-sample\) correction, which \$J = \(1 - G\)/\(1 - F\)\$ inherits.](#) The four vectors are stacked as channels of a single tensor of shape $(N, 4, 512)$, where each channel is z-scored using train-set mean and standard deviation.

Topological features. We compute the H_0 and H_1 persistence diagrams using two complementary filtrations, alpha and DTM₁₀, obtaining four diagrams per pattern. [In Section 6 we show that both are needed, and neither suffices individually.](#) Each persistence diagram is vectorized into a *persistence image*. The four persistence images are encoded separately: [a \$64 \times 64\$ image for alpha- \$H_1\$, DTM₁₀- \$H_0\$ and DTM₁₀- \$H_1\$, and a 64-dimensional vector for alpha- \$H_0\$, whose births are identically zero.](#) See Appendix C for a full treatment of the vectorization process and its hyperparameters.

Classical encoders. The classical feature’s $(N, 4, 512)$ tensor passes through a 1-D convolutional encoder: three blocks of kernel-3, padding-1 convolutions with channel widths $32 \rightarrow 64 \rightarrow 128$, each followed by ReLU; between blocks, average pooling (kernel 2) and channel-wise dropout ($p = 0.2$) downsample and regularize. Rather than pooling the final feature map to a single value per channel, we apply an adaptive average pool to a fixed width of 4. Position along r is part of the signal so, being translation-invariant, a global average would leave

[TBD: pipeline figure]

Figure 2: [TBD: pipeline figure, one row: pattern $\rightarrow$ summary curves and persistence images $\rightarrow$ encoders $\rightarrow$ head $\rightarrow (\hat{f}, \hat{\theta}) \rightarrow$ simulations $\rightarrow$ kernel score against $y_{f,\theta}$, with P^0 and P^* as references.]

the encoder unable to distinguish a feature at small r from the same feature at large r . The resulting 128×4 map is flattened and passed through a linear layer with ReLU to produce a 64-dimensional embedding.

Topological encoders. The four persistence images are encoded independently by their own copy of the same conv-pool-dropout architecture (channels $32 \rightarrow 64 \rightarrow 128$, kernel 3, dropout 0.2, adaptive pool to width 4). We use 2-D for the three images and 1-D for alpha- H_0 . Encoders are not shared across filtration or homology dimension because the persistence image axes are calibrated per-(filtration, dimension) pair, so the same pixel coordinate means something different in each. The independent encoders produce four 64-dimensional embeddings.

FCNN head. The classical and topological embeddings are concatenated with standardized $\log n$, giving a 321-dimensional input to the shared head. Two fully-connected layers (64, 32) with ReLU and dropout ($p = 0.1$) are followed by a final linear layer with either 8 logits (classification) or a scalar output for each family parameter (inference).

5 TRAINING PROCEDURE

We describe how ParamNet, and the models with which it is compared, are trained and evaluated. We evaluate the overall fidelity of the end-to-end pipeline, as well as the individual components of the pipeline independently.

Training. Every network is trained using the same recipe: AdamW with learning rate 10^{-3} , batch size 128, at most 200 epochs, early stopping on validation loss with patience 30, and the best-validation weights restored. The classifier minimizes class-weighted cross-entropy, while the parameter estimators minimize squared error on standardized $\log \theta$. Because the

recipe is shared, a difference between two networks is a difference in their inputs. Settings, parameter counts and training cost are in Appendix D.

Controls. As controls we train gradient-boosted trees on feature tables of the summary curves, of persistence summaries, and of both, and a logistic regression on the summary curves. We also retrain ParamNet with the persistence images removed, or with one filtration only, and nothing else changed (Appendix D).

Classical baseline. Our classical baseline is minimum contrast on K , with penalized-contrast model selection among Poisson and the families that have a closed-form K . Strauss and cell have none, so it can neither select nor fit them; given either, it falls back to the training median. We also run it given the true family, which removes its selection errors.

Evaluation. Every model is scored on the same held-out test patterns. For classification we report balanced accuracy over the eight families, and over the four mechanisms (CSR, clustered, regular, cell). For inference, we report the estimator error $\text{RMSE}(\log \theta)/\text{s.d.}$ averaged over a family’s targets, where 1 is the error of predicting the training mean.

5.1 Goodness of Fit

Accuracy and estimation error in terms of RMSE on log-normalized parameters does not tell us whether the fitted model reproduces the observed pattern, nor does it provide clear intuition of the model’s performance. We therefore score the test set on fidelity directly.

Let $\{\mathbf{x}_{f_1, \theta_1}, \dots, \mathbf{x}_{f_N, \theta_N}\}$ be the test patterns, with held-out replicates $\{\mathbf{y}_{f_1, \theta_1}, \dots, \mathbf{y}_{f_N, \theta_N}\}$. For test pattern $\mathbf{x}_{f_i, \theta_i}$ with ParamNet fit $\hat{P}_i = (\hat{f}_i, \hat{\theta}_i)$, write $P_i^* = (f_i, \theta_i)$ for the oracle and $P_i^0 = (\text{Poisson}, \hat{\lambda})$ for the CSR fit. Each $P_i \in \{\hat{P}_i, P_i^*, P_i^0\}$ is scored by com-

paring the held-out replicate $\mathbf{y}_{f_i, \theta_i}$ to 16 independent simulations $z_1, \dots, z_{16} \stackrel{\text{iid}}{\sim} P_i$ on W , using a kernel score on local configurations. Its properties and validation are in Appendix F.

Local configurations. For a pattern z with n points, rescale to unit intensity, $\tilde{z} = \sqrt{n}z$, and fix a centre set $C(z) \subseteq \tilde{z}$ of size $m = \min(n, 300)$, drawn uniformly at random. Each centre $p \in C(z)$ gives a local configuration

$$N_R(p, \tilde{z}) = \{q \in \tilde{z} \setminus \{p\} : \|q - p\| \leq R\}, \quad (1)$$

$$\phi(p; z) = \frac{1}{R} (N_R(p, \tilde{z}) - p), \quad (2)$$

with $R = 1.5$ (in units of mean spacings since $\tilde{z}$ has unit intensity). We write $\phi_1, \dots, \phi_m$ for the local configurations of the held out $\mathbf{y}_{f_i, \theta_i}$.

Kernel score. Local configurations are compared by an RBF kernel between Gaussian-smoothed occupancy measures on a 21×21 grid,

$$\mu_\phi(u) = \sum_{p \in \phi} e^{-\frac{\|u - p\|^2}{2h^2}}, \quad (3)$$

$$k(\phi, \psi) = e^{-\frac{\|\mu_\phi - \mu_\psi\|^2}{2\ell^2}}, \quad (4)$$

with $h = 0.1$ and bandwidth ℓ fixed to the median pairwise distance among $\mu_{\phi_1}, \dots, \mu_{\phi_m}$ (before any model is scored).

A model P_i is scored against $\mathbf{y}_{f_i, \theta_i}$ by the kernel score

$$S(P_i, \mathbf{y}_{f_i, \theta_i}) = \frac{1}{2} \mathbb{E} k(\Phi, \Phi') - \frac{1}{m} \sum_{i=1}^m \mathbb{E} k(\phi_i, \Phi), \quad (5)$$

where Φ, Φ' are local configurations $\phi(p; z_k), \phi(p'; z_{k'})$ at centres drawn from two *different* simulations $z_k, z_{k'}$ ($k \neq k'$) of P_i , with expectations estimated by averaging over the 16 simulations above.

Skill. For test set $\{\mathbf{x}_{f_1, \theta_1}, \dots, \mathbf{x}_{f_N, \theta_N}\}$, skill sums (over $i = 1, \dots, N$) how each fit's kernel score on y_{f_i, θ_i} compares to P_i^0 and P_i^* (reported only when the denominator is resolved).

$$\text{skill} = \frac{\sum_i [S(P_i^0, y_{f_i, \theta_i}) - S(\hat{P}_i, y_{f_i, \theta_i})]}{\sum_i [S(P_i^0, y_{f_i, \theta_i}) - S(P_i^*, y_{f_i, \theta_i})]} \quad (6)$$

Skill 0 means no better than CSR and skill 1 as good as the truth; on a finite set of patterns a fit can score slightly above 1. A fit's *gain over CSR* on pattern i is $S(P_i^0, y_i) - S(P, y_i)$, the amount by which it improves on the CSR reference; skill weights each pattern by the true model's gain, so patterns on which no model beats CSR carry no weight.

Evaluation set. We score fits on 1 440 test patterns: 60 per regime stratum (below the boundary at $\tau = 0.5$, between $\tau = 0.5$ and 0.9, and beyond $\tau = 0.9$; Section 6) for each family with a coordinate, 60 per band of k for cell ($k = 2, 3-4, \geq 5$), and 180 Poisson patterns (1 439 scored: one ring pattern's gain could not be resolved). Intervals are 95% bootstrap intervals over patterns, and every difference is paired: same patterns, same simulations.

6 EXPERIMENTS ON SIMULATED PATTERNS

We score ParamNet end-to-end on the 1 440 patterns of the evaluation set, and its classifier on all 192 000 test patterns (Section 5.1). We find that ParamNet's fits reproduce held-out data nearly as well as the true model, and better than minimum contrast. The model's classification accuracy is set by the data, not by the model. Moreover, ParamNet's classification errors cost almost no fidelity.

High fidelity fits. ParamNet reaches skill 0.90 [0.85, 0.95]: its fits recover 90% of what the true model gains over CSR. By comparison, minimum contrast reaches 0.74 [0.65, 0.81] on the same patterns, [a paired gap of +0.16 \[+0.08, +0.26\], and +0.11 \[+0.04, +0.21\] when both are given the true family](#). ParamNet's advantage is concentrated in the Strauss and cell families, which have no closed-form K . On these two families ParamNet reaches skill 0.93 [0.79, 1.06], while minimum contrast, even when given the true family, does no better than CSR (-0.04 [$-0.17, +0.08$]; per family: Table 7).

ParamNet's advantage is also visible in the estimators' loss: given the true family, every learned estimator has lower RMSE($\log \theta$)/s.d. than minimum contrast on every family. On the Neyman–Scott families minimum contrast does worse than predicting the training mean (1.48–2.25; Appendix H.3).

Classification difficulty is inherent. The deliberately wide priors admit many near-CSR patterns, on which every method struggles to identify the family. We call a pattern *detectable* if a test of CSR rejects it. Every test we use is calibrated to falsely reject only 5% of truly Poisson patterns, so rejection is real evidence of structure. For each family, the share of patterns detected is governed by one physical coordinate (Table 1), so the undetectable region is bounded by a single curve in parameter space (Figure 3). The τ -boundary is the curve along which a fraction τ of a family's patterns are detected; patterns beyond it are detected more often. We fit it on a reference de-

Table 2: Main results. *Identification*: balanced accuracy over the eight families on the 192 000 test patterns, on all of them and in the structured regime ($\tau=0.9$), and over the four mechanisms (CSR, clustered, regular, cell) at $\tau=0.9$; intervals are within ± 0.003 . Minimum contrast never selects Strauss or cell. *Fidelity*: skill on the 1 439 scored patterns (1: as good as the true model; 0: no better than CSR), given the predicted family, given the true family, and on the non-Poisson patterns routed to a wrong family. Δ : ParamNet – minimum contrast, paired, overall and by whether minimum contrast can fit the family. Intervals are 95% bootstrap over patterns; higher is better. Accuracy is modest, yet the fits recover 90% of the true model’s gain.

	Identification (balanced accuracy)			Fidelity (skill)		
	All	$\tau=0.9$	Mechanism	Predicted family	True family	Wrong family (n)
ParamNet	0.450	0.754	0.899	0.90 [0.85, 0.95]	0.89 [0.84, 0.94]	0.82 [0.73, 0.90] (350)
Minimum contrast	0.255	0.408	0.633	0.74 [0.65, 0.81]	0.78 [0.65, 0.87]	0.57 [0.40, 0.70] (647)
Δ , all families				+0.16 [+0.08, +0.26]	+0.11 [+0.04, +0.21]	
Δ , closed-form K				+0.06 [−0.00, +0.15]	+0.04 [−0.03, +0.14]	
Δ , Strauss, Cell				+1.27 [+0.71, +2.04]	+0.97 [+0.80, +1.17]	

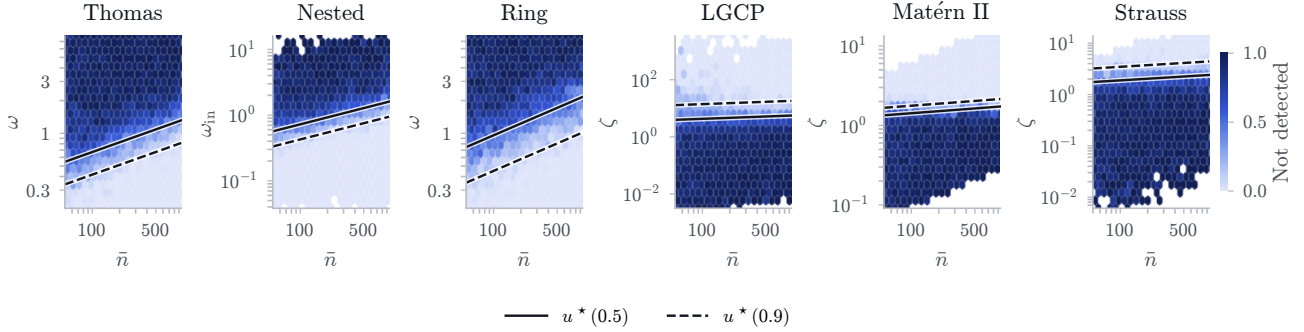


Figure 3: The regime boundary in the physical parameters: share of test patterns the reference detector (trees on the summary curves, size 5%) does not detect, over $\bar{n}$ and each family’s regime coordinate η (Table 1). Each boundary is a line $u = \log \eta + a \log \bar{n} = u^*(\tau)$: $\tau = 0.5$ solid, 0.9 dashed. Patterns approach CSR as ω grows or ζ shrinks.

tector, trees on the summary curves (Appendix G). Detectability is a property of the patterns, not of the test. The L test, trees on persistence summaries alone and ParamNet, thresholded to the same false-positive rate, detect the same share of every family’s patterns to within three points, and meet the same boundary (Figure 4; Table 5). Restricting to patterns beyond the $\tau = 0.5$ boundary raises ParamNet’s accuracy from 0.45 to 0.70, and beyond $\tau = 0.9$ to 0.75 (per learner: Table 6). Below the $\tau = 0.5$ boundary ParamNet calls 61% of patterns Poisson, and even the true model gains nothing over CSR (Figure 5).

Errors cost almost no fidelity. ParamNet makes two kinds of error, and we show that neither result in significant loss of fidelity. Supplying ParamNet with the true family changes its skill by -0.01 $[-0.03, +0.02]$, implying that patterns that get misclassified in the end-to-end pipeline still receive a high-fidelity fit. In fact, wrong-family fits keep 82% of the true model’s gain (against 91% when the family is identified), be-

cause the confusions occur between look-alikes: in the structured regime 96% stay within one mechanism (clustered, regular, cell; Figure 6), and three quarters are among Thomas, nested and ring, which produce the same patterns over part of their parameter spaces (Appendix A). Calling a pattern Poisson costs nothing: on the 271 patterns ParamNet sends to Poisson, the true model does no better than CSR, and the L test rejects CSR on only 2.2% [1.0, 4.7] of them, no more often than on Poisson patterns. For these patterns, CSR is essentially the correct fit.

Topology helps where minimum contrast cannot fit. To investigate the value of each filtration, we retrain ParamNet on alpha alone and on DTM_{10} alone. Table 3 shows that the two are complementary. Alpha alone cuts estimation error on Strauss by 28%, as much as both filtrations together (26%), while DTM_{10} alone does the same for cell (19% against 18%). Neither does both: alpha gets only 11% on cell and DTM_{10} only 10% on Strauss, which is why ParamNet uses

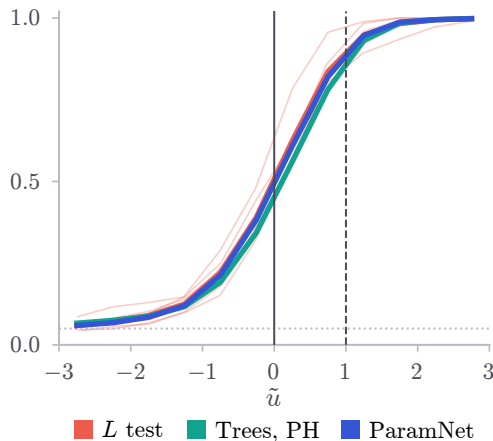


Figure 4: Detection power of three tests of size 5% (dotted) against the rescaled regime coordinate $\tilde{u}$ (Figure 5), families averaged; faint: the L test per family. Solid: the $\tau = 0.5$ boundary; dashed: $\tau = 0.9$. The boundaries are fitted on a fourth detector, so the agreement is not by construction.

both. Together, the persistence images cut error by 26% on Strauss and 18% on cell, the two families that minimum contrast cannot fit.

Summary. (i) ParamNet’s fits reach skill 0.90, against 0.74 for minimum contrast; (ii) the data, not the model, limits accuracy: every detector meets the same boundary, and below it even the true model gains nothing over CSR; (iii) errors are cheap: supplying the true family changes skill by -0.01 , and calling near-CSR patterns Poisson costs nothing; (iv) topology helps where minimum contrast fails, cutting estimation error on Strauss and cell by 26% and 18%.

Table 3: Relative change in estimation error $\text{RMSE}(\log \theta)/\text{s.d.}$ ($\tau=0.9$) when persistence images are added to the same network on the summary curves (“Curves alone”: its error). Negative is better; one training seed (intervals: Appendix H.4). The images help where minimum contrast cannot fit.

	Curves alone	+ both	+ alpha	+ DTM
Strauss	0.152	−26%	−28%	−10%
Cell	0.173	−18%	−11%	−19%
Thomas	0.415	−2%	+3%	+3%
Nested	0.576	−1%	−1%	−1%
Ring	0.457	+1%	−2%	0%
LGCP	0.301	−4%	−2%	−2%
Matérn II	0.084	+7%	+6%	+41%

Supplementary Materials

A FAMILIES

- Definitions, one paragraph each; trim the Poisson paragraph.
- Closed-form K for the families that have one. (new; minimum contrast depends on them)
- Regime coordinates, one-line rationale each. (moved from Appendix G)
- Near-equivalences between families: ring $\rightarrow$ Thomas as $\rho/\sigma \rightarrow 0$; nested $\rightarrow$ Thomas for large σ_1 ; Matérn II $\approx$ Strauss as $q \rightarrow 1$.

B SIMULATION

- Prior table: family, sampled coordinates, ranges, solved parameter. Replaces the current bullets.
- Constraints: ≥ 4 effective clusters; conditioning on $n \in [20, 2000]$ (range of the L test’s null tables; 304 of 800 304 simulations redrawn); parent buffer.
- Simulators: Strauss MCMC (sweeps, convergence check), cell, Matérn II, LGCP field method and grid. (LGCP new)
- Volume; split by θ .
- Cut the “minimal curation” bullets; one sentence in Section 3 covers them.

C FEATURES

- Summary curves: estimators, edge corrections, r grid, standardisation. (move the edge-correction sentence from Section 4)
- Filtrations: alpha and DTM_k definitions, $k = 10$, software. (new; neither is defined anywhere)
- Persistence images: weighting, bandwidth, boxes, scaling by $\sqrt{n}$ and n , $\sqrt{\cdot}$ transform. (text in the backup; drop the Rips mention)
- Feature tables used by the trees. (new)
- Optional figure: mean persistence image per family ($\text{alpha-}H_1$, $\text{DTM-}H_0$).

D MODELS AND TRAINING

- Layer-by-layer architecture table. (move the encoder paragraphs from Section 4; saves about 0.25 page)
- Training recipe, class weights, target standardisation. (text in the backup)
- Tree and logistic hyperparameters. (new)
- Parameter counts, hardware, wall clock (Table 4); shorten the caption.
- Training-seed variation of the curves vs curves+PH comparison. (blocked: seedcheck)

Table 4: Each model’s input representation and training cost. “Classifier” is wall-clock seconds to train the one 8-way classifier until early stopping; “Estimators” is the same summed over the seven per-family estimators. “Input” names the tensors the model sees: *curves* is the (4,512) summary-function stack, *classical* the same curves as a feature table for the tree learners, and each filtration name a persistence summary of that filtration’s diagrams. All four network rows share one trainer and stopping rule, so their differences are the cost of the persistence images alone. Lower is cheaper; cost is not the point of the comparison.

Model	Learner	Input	Classifier (s)	Estimators (s)
<code>fusion_curves_alpha_dtm10</code>	nn	curves, alpha_diameter, dtm_k10	3,217	7,090
<code>hgb_classical</code>	hgb	classical	414	132
<code>hgb_classical_ph</code>	hgb	classical, ph_alpha, ph_dtm10	188	218
<code>hgb_ph</code>	hgb	ph_alpha, ph_dtm10	153	—
<code>linear_classical</code>	linear	classical	90	—
<code>nn_curves</code>	nn	curves	1,862	604
<code>fusion_curves_alpha</code>	nn	curves, alpha_diameter	1,776	1,680
<code>fusion_curves_dtm10</code>	nn	curves, dtm_k10	2,081	1,744

E MINIMUM-CONTRAST BASELINE

- Contrast: K , r range, exponents, optimizer, bounds, starting values. (new)
- Penalised selection: penalty, how it was tuned, on which split.
- Fallback to the training median for Strauss and cell; failure rate.
- Strauss has a standard maximum-pseudolikelihood fit: justify leaving it out, or add it as a baseline.

F KERNEL SCORE

- Propriety: state with a short proof (MMD identity), or drop “Proposition” and cite.
- Fixed choices: R , h , grid, ℓ , m , edge rule, 16 simulations.
- Skill estimator: ratio of sums, paired bootstrap, the “resolved denominator” rule, the one excluded ring pattern.
- Wrong-model ladder. (blocked: dose run on v2; v1 numbers cannot stay)
- Monte Carlo repeatability. (blocked: fidelity seeds 1–4)

G REGIME BOUNDARY

- Detectors: L -test statistic and calibration; 5% thresholds for the learned detectors; false-positive rates 0.045–0.053.
- Fit of a and u^* ; AUC of u vs all parameters; agreement across detectors (Table 5).
- Per-family detection and gain, cell against k (Figure 5).
- Boundary in the physical parameters (Figure 3).
- Gain over CSR per stratum and family. (moved from the old power subsection; now in Figure 5)
- Construction of the evaluation strata.

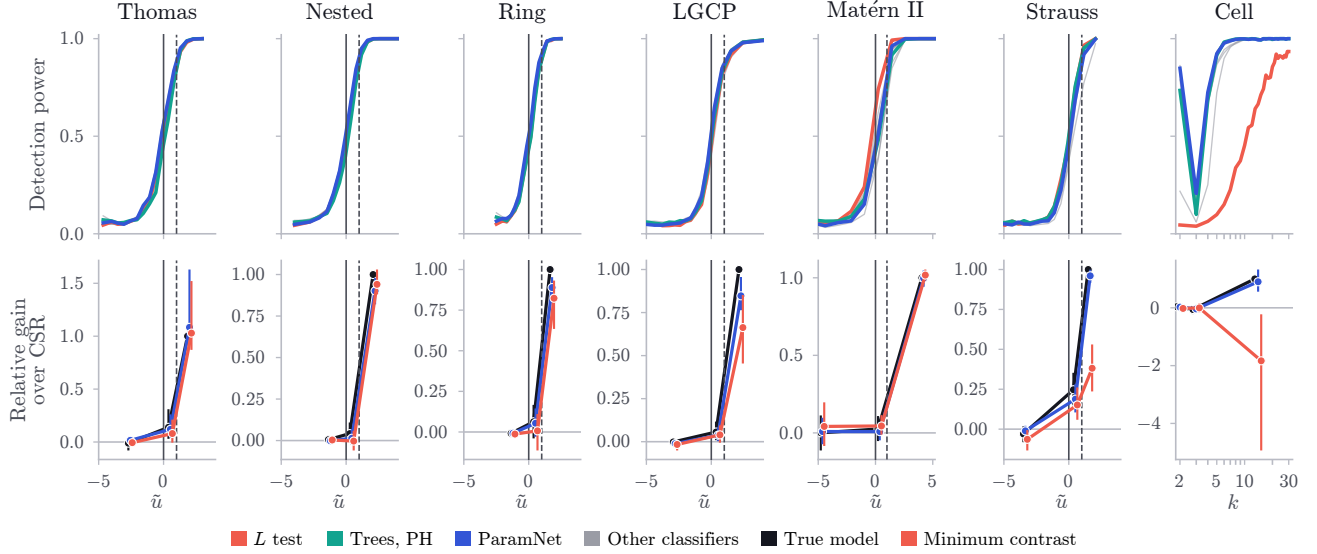


Figure 5: Figure 4 per family, cell against k . Each family’s coordinate $u = \log \eta + a \log \bar{n}$ is rescaled to $\tilde{u} = (u - u_{0.5}^*) / (u_{0.9}^* - u_{0.5}^*)$, so that the $\tau = 0.5$ boundary is at $\tilde{u} = 0$ (solid) and the $\tau = 0.9$ boundary at $\tilde{u} = 1$ (dashed). Top: detection power on the test patterns of the L test, trees on persistence summaries alone and ParamNet, and in grey every other classifier, all tests of size 5%. Bottom: gain over CSR of the true model, ParamNet and minimum contrast on each stratum of the evaluation set, as a share of the true model’s gain in the family’s strongest stratum (cell: $k \geq 5$); 95% bootstrap intervals over patterns. The L test detects cell at its nominal 5% for $k \leq 4$ and in 21% of patterns at $k = 5-9$, where the learned detectors already detect almost all. Minimum contrast never selects cell, and its fits to cell patterns at $k \geq 5$ do worse than CSR.

Table 5: The regime boundary per family; regime coordinates η as in Table 1. Detection is a test of size 5%; $u = \log \eta + a \log \bar{n}$. AUC: detection predicted from u alone and from all parameters (boosted trees, out of fold). $u^*(0.5)$: the boundary, where detection power reaches 0.5, fitted on the reference detector’s out-of-fold training predictions, and recomputed on the test split for the L test, for trees on persistence summaries alone (“PH only”, which never sees L) and for ParamNet (95% bootstrap intervals over θ). Last column: share of each family’s patterns in the structured regime. Higher AUC is better. The four boundary estimates agree to within 0.25 in u in every family. Cell has no monotone coordinate. The L test detects it at its nominal 5% for $k \leq 4$, while trees on the summary curves already detect 84% of patterns at $k = 2$ (ParamNet 85%): the nearest-neighbour summaries see what K cannot.

Family	AUC		Boundary $u^*(0.5)$				In regime $\tau = 0.5 / 0.9$
	a	u / all	Fitted	L test	PH only	ParamNet	
Thomas	-0.30	0.935 / 0.940	-1.77	-1.71 [-1.73, -1.68]	-1.81 [-1.88, -1.77]	-1.71 [-1.74, -1.61]	0.39 / 0.26
Nested	-0.35	0.950 / 0.959	-1.95	-1.95 [-2.02, -1.92]	-2.04 [-2.09, -1.98]	-1.95 [-2.01, -1.86]	0.58 / 0.43
Ring	-0.36	0.928 / 0.945	-1.70	-1.82 [-1.86, -1.77]	-1.84 [-1.89, -1.77]	-1.65 [-1.79, -1.65]	0.50 / 0.30
LGCP	-0.12	0.927 / 0.936	+0.91	+0.95 [+0.92, +1.17]	+0.91 [+0.87, +1.00]	+0.70 [+0.69, +0.92]	0.31 / 0.21
Matérn II	-0.09	0.957 / 0.962	-0.05	-0.09 [-0.10, -0.05]	+0.01 [-0.00, +0.06]	+0.02 [+0.00, +0.03]	0.42 / 0.37
Strauss	-0.11	0.904 / 0.914	+0.14	+0.18 [+0.13, +0.22]	+0.14 [+0.13, +0.18]	+0.16 [+0.13, +0.22]	0.21 / 0.11

H ADDITIONAL RESULTS

H.1 Classification by Learner

- Per-learner accuracy (Table 6); confusion matrix (Figure 6).

Table 6: Balanced accuracy over the eight families on the 192 000 test patterns. “All” is every pattern; “ $\tau=0.5$ ” and “ $\tau=0.9$ ” restrict to the structured regime at that detection power (Section 6); “Coarse” is accuracy over the four mechanisms (CSR, clustered, regular, cell) at $\tau=0.9$. Learners are named by input: “classical” is the summary curves, “PH” the persistence summaries of the alpha and DTM_{10} diagrams. Minimum contrast is penalized-contrast selection, which never selects Strauss or cell. Intervals are within ± 0.003 ; higher is better throughout. Every learner meets the same limit on all patterns, and ParamNet is ahead in the structured regime.

Classifier	All	$\tau=0.5$	$\tau=0.9$	Coarse
Trees, classical + PH	0.452	0.683	0.735	0.862
ParamNet	0.450	0.698	0.754	0.899
Trees, classical	0.447	0.675	0.726	0.858
ParamNet, classical	0.446	0.691	0.747	0.889
Trees, PH	0.436	0.665	0.719	0.849
Logistic, classical	0.396	0.583	0.620	0.783
Minimum contrast	0.255	0.382	0.408	0.633

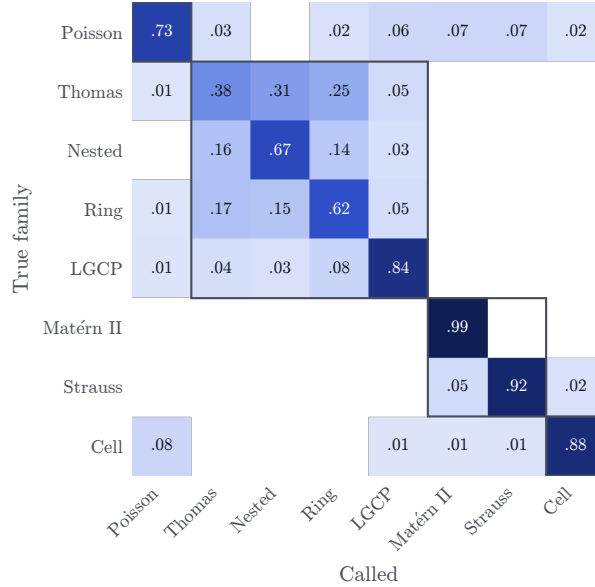


Figure 6: Confusion matrix of ParamNet in the structured regime ($\tau = 0.9$), rows normalized: the share of each true family assigned to each family; blank cells are below 0.005. Boxes: the mechanisms (CSR, clustered, regular, cell). Confusions stay within the mechanism, mostly among Thomas, nested and ring, which produce the same patterns over part of their parameter spaces.

H.2 Per-Family Fidelity

- Per-family skill (Table 7). (blocked: 5-seed rescaling)

H.3 Per-Family Estimation Error

- Table 8.

H.4 Persistent Homology by Filtration

- Filtration contrast (Table 9).
- 40-input ablation only if it lands, as one heatmap; otherwise remove the promise.

Table 7: Skill per family, 180 patterns each (179 for ring). “ Δ ” is ParamNet – minimum contrast on the same patterns, given the predicted and the true family, so minimum contrast’s own skill is ParamNet’s minus Δ . “ $\dagger$ ” marks the families with no closed-form K : minimum contrast never selects them, and given the true family it falls back to the training median. Skill can exceed 1, because on a single replicate a fit can score better than the true model. Intervals are 95% bootstrap over patterns; higher skill and positive Δ favour ParamNet.

Family	ParamNet	Δ predicted	Δ true family
Thomas	1.08 [0.90, 1.60]	+0.10 [+0.01, +0.26]	+0.08 [+0.00, +0.21]
Nested	0.87 [0.79, 0.94]	−0.03 [−0.12, +0.06]	+0.08 [−0.08, +0.34]
Ring	0.89 [0.82, 0.96]	+0.11 [−0.01, +0.29]	+0.05 [−0.05, +0.18]
LGCP	0.83 [0.74, 0.92]	+0.18 [−0.02, +0.38]	−0.05 [−0.14, +0.03]
Matérn II	0.99 [0.87, 1.12]	−0.09 [−0.25, +0.04]	−0.12 [−0.28, −0.01]
Strauss [†]	0.94 [0.87, 1.00]	+0.55 [+0.40, +0.70]	+0.86 [+0.75, +0.97]
Cell [†]	0.91 [0.50, 1.34]	+2.81 [+1.13, +6.20]	+1.22 [+0.78, +2.14]

H.5 Other

- Gallery of observed patterns vs fits: random draws per family, including wrong-family fits.
- Matérn I out-of-library result only if Limitations cites it; otherwise cut.

Table 8: Estimation error per family and estimator, given the true family. Cells are $\text{RMSE}(\log \theta)/\text{s.d.}$ averaged over the family’s targets, where 1 is the error of predicting the training mean. “Val” is the validation split; “All” is every test pattern; “ $\tau=0.5$ ” and “ $\tau=0.9$ ” restrict to the structured regime at that detection power. “ $\star$ ” marks the lowest validation error, the estimator a per-family selection rule would pick. Intervals are 95% over test θ ; lower is better. Every learner beats minimum contrast on every family.

Family	Estimator ($\star$ = best on val)	Val	All	$\tau=0.5$	$\tau=0.9$
Thomas	fusion_curves_alpha_dtm10 $\star$	0.742	0.739 [0.733, 0.746]	0.548 [0.538, 0.558]	0.407 [0.397, 0.417]
Thomas	fusion_curves_dtm10	0.743	0.741 [0.734, 0.748]	0.559 [0.549, 0.569]	0.426 [0.416, 0.437]
Thomas	fusion_curves_alpha	0.745	0.741 [0.735, 0.748]	0.556 [0.546, 0.566]	0.427 [0.416, 0.438]
Thomas	nn_curves	0.746	0.741 [0.735, 0.748]	0.553 [0.543, 0.563]	0.415 [0.404, 0.426]
Thomas	hgb_classical_ph	0.750	0.744 [0.738, 0.750]	0.553 [0.542, 0.563]	0.421 [0.409, 0.430]
Thomas	hgb_classical	0.752	0.745 [0.739, 0.751]	0.555 [0.545, 0.565]	0.425 [0.414, 0.435]
Thomas	min. contrast (tuned)	—	1.762	—	—
Nested	fusion_curves_alpha	0.729	0.717 [0.711, 0.724]	0.638 [0.631, 0.646]	0.570 [0.564, 0.577]
Nested	fusion_curves_dtm10 $\star$	0.728	0.717 [0.712, 0.725]	0.636 [0.629, 0.644]	0.571 [0.563, 0.578]
Nested	fusion_curves_alpha_dtm10	0.729	0.719 [0.714, 0.726]	0.639 [0.631, 0.647]	0.571 [0.564, 0.578]
Nested	nn_curves	0.731	0.719 [0.714, 0.726]	0.642 [0.635, 0.649]	0.576 [0.569, 0.583]
Nested	hgb_classical_ph	0.735	0.724 [0.719, 0.731]	0.641 [0.634, 0.649]	0.573 [0.565, 0.580]
Nested	hgb_classical	0.740	0.726 [0.721, 0.733]	0.647 [0.641, 0.655]	0.580 [0.572, 0.587]
Nested	min. contrast (tuned)	—	2.250	—	—
LGCP	hgb_classical	0.542	0.541 [0.536, 0.545]	0.356 [0.348, 0.365]	0.279 [0.269, 0.289]
LGCP	fusion_curves_alpha_dtm10 $\star$	0.540	0.541 [0.536, 0.546]	0.364 [0.355, 0.372]	0.289 [0.280, 0.299]
LGCP	nn_curves	0.542	0.541 [0.536, 0.546]	0.366 [0.357, 0.374]	0.301 [0.290, 0.311]
LGCP	hgb_classical_ph	0.543	0.541 [0.537, 0.546]	0.359 [0.350, 0.367]	0.283 [0.273, 0.293]
LGCP	fusion_curves_alpha	0.541	0.541 [0.537, 0.546]	0.365 [0.356, 0.373]	0.293 [0.283, 0.303]
LGCP	fusion_curves_dtm10	0.544	0.542 [0.537, 0.547]	0.363 [0.354, 0.371]	0.295 [0.284, 0.306]
LGCP	min. contrast (tuned)	—	0.873	—	—
Matérn II	fusion_curves_alpha	0.238	0.233 [0.230, 0.236]	0.114 [0.111, 0.117]	0.089 [0.087, 0.092]
Matérn II	nn_curves	0.240	0.233 [0.230, 0.236]	0.103 [0.101, 0.105]	0.084 [0.082, 0.086]
Matérn II	hgb_classical_ph $\star$	0.238	0.234 [0.231, 0.237]	0.109 [0.106, 0.112]	0.088 [0.086, 0.090]
Matérn II	hgb_classical	0.240	0.235 [0.232, 0.238]	0.110 [0.108, 0.113]	0.089 [0.086, 0.092]
Matérn II	fusion_curves_alpha_dtm10	0.241	0.236 [0.233, 0.239]	0.116 [0.113, 0.119]	0.090 [0.088, 0.093]
Matérn II	fusion_curves_dtm10	0.245	0.243 [0.240, 0.246]	0.137 [0.134, 0.140]	0.118 [0.116, 0.121]
Matérn II	min. contrast (tuned)	—	0.315	—	—
Ring	fusion_curves_dtm10	0.706	0.709 [0.702, 0.716]	0.601 [0.591, 0.611]	0.457 [0.448, 0.466]
Ring	fusion_curves_alpha $\star$	0.703	0.709 [0.702, 0.716]	0.599 [0.590, 0.607]	0.449 [0.439, 0.459]
Ring	fusion_curves_alpha_dtm10	0.705	0.710 [0.703, 0.717]	0.599 [0.590, 0.608]	0.461 [0.451, 0.470]
Ring	hgb_classical_ph	0.705	0.710 [0.703, 0.717]	0.590 [0.581, 0.600]	0.436 [0.428, 0.445]
Ring	nn_curves	0.707	0.711 [0.704, 0.718]	0.605 [0.596, 0.614]	0.457 [0.449, 0.466]
Ring	hgb_classical	0.715	0.715 [0.708, 0.722]	0.604 [0.595, 0.613]	0.462 [0.453, 0.471]
Ring	min. contrast (tuned)	—	1.475	—	—
Cell	hgb_classical_ph $\star$	0.132	0.137 [0.135, 0.140]	0.137 [0.135, 0.140]	0.137 [0.135, 0.140]
Cell	fusion_curves_dtm10	0.137	0.141 [0.139, 0.143]	0.141 [0.139, 0.143]	0.141 [0.139, 0.143]
Cell	fusion_curves_alpha_dtm10	0.138	0.143 [0.140, 0.145]	0.143 [0.140, 0.145]	0.143 [0.140, 0.145]
Cell	fusion_curves_alpha	0.148	0.153 [0.151, 0.156]	0.153 [0.151, 0.156]	0.153 [0.151, 0.156]
Cell	hgb_classical	0.154	0.159 [0.157, 0.161]	0.159 [0.157, 0.161]	0.159 [0.157, 0.161]
Cell	nn_curves	0.172	0.173 [0.171, 0.176]	0.173 [0.171, 0.176]	0.173 [0.171, 0.176]
Cell	min. contrast (tuned)	—	0.547	—	—
Strauss	hgb_classical $\star$	0.534	0.546 [0.537, 0.552]	0.203 [0.197, 0.212]	0.102 [0.097, 0.108]
Strauss	hgb_classical_ph	0.538	0.546 [0.538, 0.552]	0.203 [0.195, 0.211]	0.099 [0.095, 0.105]
Strauss	fusion_curves_alpha	0.543	0.555 [0.547, 0.561]	0.226 [0.218, 0.234]	0.110 [0.104, 0.115]
Strauss	nn_curves	0.544	0.555 [0.547, 0.561]	0.262 [0.255, 0.270]	0.152 [0.146, 0.157]
Strauss	fusion_curves_alpha_dtm10	0.552	0.562 [0.555, 0.568]	0.225 [0.218, 0.234]	0.113 [0.106, 0.119]
Strauss	fusion_curves_dtm10	0.556	0.565 [0.558, 0.571]	0.251 [0.243, 0.258]	0.137 [0.131, 0.142]
Strauss	min. contrast (tuned)	—	0.664	—	—

Table 9: Which filtration carries what topology adds. Columns are the same network, trainer and stopping rule on four input sets: “both” persistence filtrations (ParamNet), “alpha” or “DTM” alone, and “neither” (the summary curves only). Cells are $\text{RMSE}(\log \theta)/\text{s.d.}$ in the structured regime ($\tau=0.9$), where 1 is the error of predicting the training mean, then balanced accuracy at $\tau=0.9$, then ParamNet’s end-to-end skill minus each column’s, paired on the same 1439 patterns. Intervals are 95% over test θ (errors, accuracy) or over patterns (skill). One training seed, so differences under about 0.005 in error are within seed noise.

	both	alpha	DTM	neither
Thomas	0.407 [0.397, 0.417]	0.427 [0.416, 0.438]	0.426 [0.416, 0.437]	0.415 [0.404, 0.426]
Nested	0.571 [0.564, 0.578]	0.570 [0.564, 0.577]	0.571 [0.563, 0.578]	0.576 [0.569, 0.583]
LGCP	0.289 [0.280, 0.299]	0.293 [0.283, 0.303]	0.295 [0.284, 0.306]	0.301 [0.290, 0.311]
Matérn II	0.090 [0.088, 0.093]	0.089 [0.087, 0.092]	0.118 [0.116, 0.121]	0.084 [0.082, 0.086]
Ring	0.461 [0.451, 0.470]	0.449 [0.439, 0.459]	0.457 [0.448, 0.466]	0.457 [0.449, 0.466]
Cell	0.143 [0.140, 0.145]	0.153 [0.151, 0.156]	0.141 [0.139, 0.143]	0.173 [0.171, 0.176]
Strauss	0.113 [0.106, 0.119]	0.110 [0.104, 0.115]	0.137 [0.131, 0.142]	0.152 [0.146, 0.157]
accuracy, $\tau=0.9$	0.754 [0.752, 0.759]	0.749 [0.746, 0.753]	0.746 [0.742, 0.750]	0.747 [0.744, 0.750]
skill, both – column	—	-0.016 [-0.055, +0.019]	-0.003 [-0.048, +0.046]	+0.011 [-0.033, +0.053]